

Functional Requirements

1. The system shall let the user start the program by typing the BasicML filename after the program name on the command line.
2. The system shall check that the user gave exactly one filename, and stop with an error if they didn't.
3. The system shall print an error and quit if the file given can't be opened.
4. The system shall read the BasicML file line by line and load each instruction into memory, starting at address 00.
5. The system shall reject any line that isn't a valid four-digit signed instruction and tell the user which line was bad.
6. The system shall store every instruction in memory as a signed four-digit number.
7. The system shall run instructions in order using a program counter that starts at 00.
8. The system shall ask the user to type in a number when it hits a READ instruction (opcode 10).
9. The system shall make sure the number the user types for READ is a whole number between -9999 and 9999.
10. The system shall print the value at a given memory spot to the screen when it hits a WRITE instruction (opcode 11).
11. The system shall add the value at a memory address to the accumulator when it hits an ADD instruction (opcode 30).
12. The system shall subtract the value at a memory address from the accumulator when it hits a SUBTRACT instruction (opcode 31).
13. The system shall multiply the accumulator by the value at a memory address when it hits a MULTIPLY instruction (opcode 33).
14. The system shall divide the accumulator by the value at a memory address when it hits a DIVIDE instruction (opcode 32), but only after checking the divisor isn't zero.
15. The system shall jump the program counter to a new address when it hits a BRANCH instruction (opcode 40).
16. The system shall stop running the program when it hits a HALT instruction (opcode 43).

Non-Functional Requirements

1. The system shall always show numbers as signed, four-digit values so the output looks consistent.
2. The system shall process each instruction fast enough that there's no noticeable delay to the user.

3. The system shall run using just Python 3, without needing the user to install any extra packages.